



College of

EDUCATION

MULTIVARIATE LINEAR MODELS (WITH LAVAAN)

- Multivariate
 - Starting with multivariate regression...
 - ...then arriving at our final destination

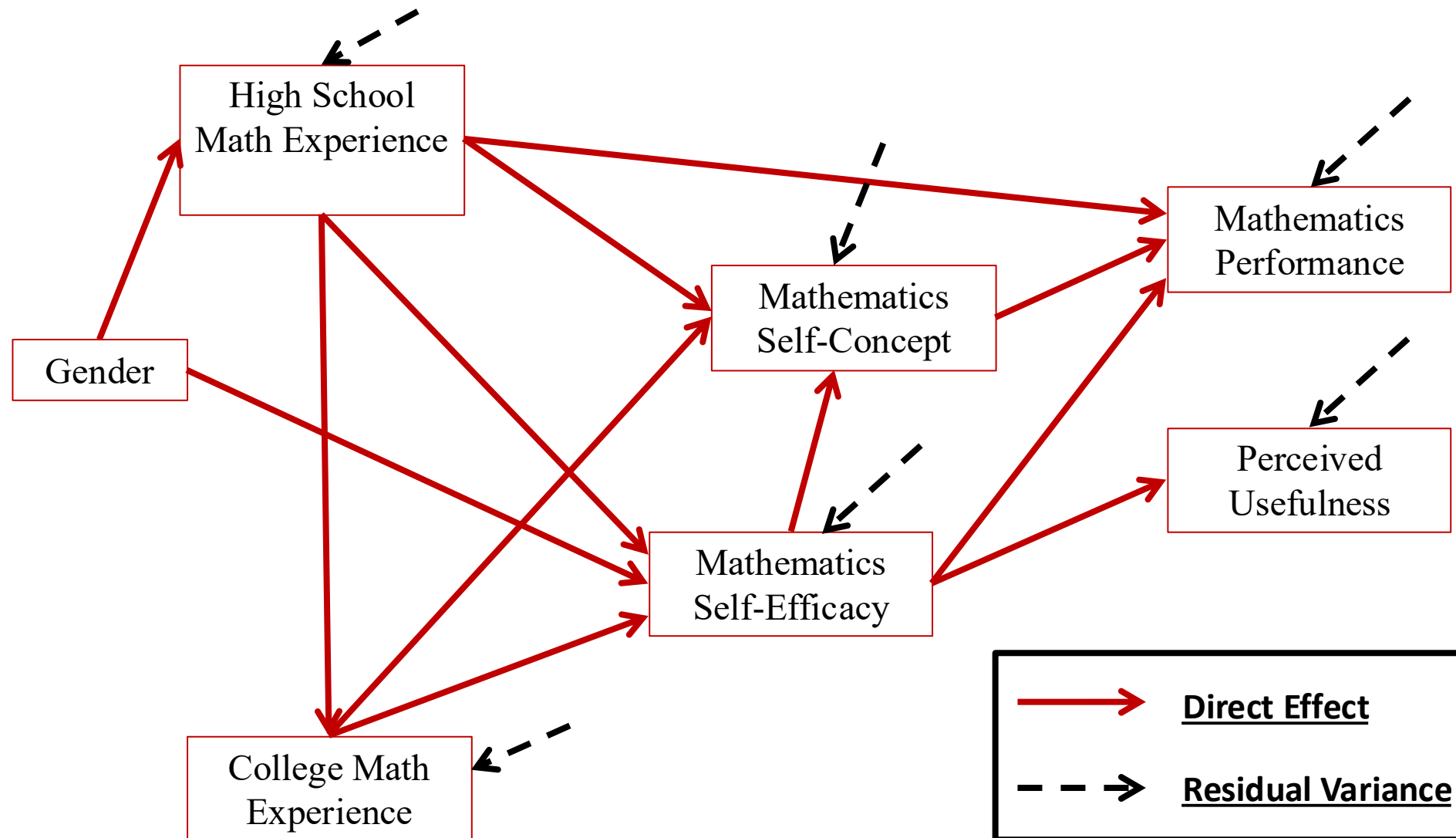
- Path analysis details:
 - Standardized coefficients
 - Model modification
 - Direct and indirect effects

- Additional issues in path analysis
 - Estimation types
 - Variable considerations

- Data are simulated based on the results reported in:
Pajares, F., & Miller, M. D. (1994). Role of self-efficacy and self-concept beliefs in mathematical problem solving: a path analysis. *Journal of Educational Psychology*, 86, 193-203.
- Sample of 350 undergraduates (229 women, 121 men)
 - In simulation, 10% of variables were missing (using missing completely at random mechanism)
- Note: simulated data characteristics differ from actual data (some variables extend beyond their official range)
 - Simulated using Multivariate Normal Distribution
 - Some variables had boundaries that simulated data exceeded
 - Results will not match exactly due to missing data and boundaries

- Gender (1 = male; 0 = female)
- Math Self-Efficacy (MSE)
 - Reported reliability of .91
 - Assesses math confidence of college students
- Perceived Usefulness of Mathematics (USE)
 - Reported reliability of .93
- Math Anxiety (MAS)
 - Reported reliability ranging from .86 to .90
- Math Self-Concept (MSC)
 - Reported reliability of .93 to .95
- Prior Experience at High School Level (HSL)
 - Self report of number of years of high school during which students took mathematics courses
- Prior Experience at College Level (CC)
 - Self report of courses taken at college level
- Math Performance (PERF)
 - Reported reliability of .788
 - 18-item multiple choice instrument (total of correct responses)

Our Destination: A Path Model



- For our class today, our path analyses assumes the variables in our analysis are multivariate normally distributed
 - Mean vectors
 - Covariance matrices
- By specifying simultaneous regression equations (the core of path models), a very specific covariance matrix is implied
- Much like MANOVA and multilevel models, the key to path analysis is finding an effective approximation to the unstructured (saturated) covariance matrix
 - With fewer parameters, if possible
- The art to path analysis is in specifying models that blend theory and statistical evidence to produce generalizable results

MULTIVARIATE REGRESSION

- We will now simultaneously model two variables from our example data that we wish to describe:
 - Mathematics performance (PERF)
 - Perceived usefulness (PERF)
- We will assume these to be continuous variables
- Initially, we will only look at an empty model with these two variables
 - Empty models are baseline models
 - We will use these to show how such models look based on the characteristics of the multivariate normal distribution
 - We will also show the bigger picture when modeling multivariate data: how we must be sure to model the covariance matrix correctly

- The multivariate model for PERF and USE is given by two regression models, which are estimated simultaneously:

$$\begin{aligned} PERF_i &= \beta_0^{PERF} + e_i^{PERF} \\ USE_i &= \beta_0^{USE} + e_i^{USE} \end{aligned}$$

- As there are two variables, the error terms have a joint distribution that will be a multivariate normal:

$$\begin{bmatrix} e_i^{PERF} \\ e_i^{USE} \end{bmatrix} \sim N_2 \left(\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{R} = \begin{bmatrix} \sigma_{e:PERF}^2 & \sigma_{e:PERF,USE} \\ \sigma_{e:PERF,USE} & \sigma_{e:USE}^2 \end{bmatrix} \right)$$

- Each error term has its own variance but now there is a covariance between error terms
 - We will soon see that the overall $\mathbf{R}$ matrix structure can be modified

- Before showing the syntax and the results, we must first describe how the multivariate empty model implies how our data should look
 - This will be true for this lecture...soon we will have Y show up on either side of the equals sign which changes the math a little

- Multivariate model with matrices:

$$\begin{bmatrix} PERF_i \\ USE_i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_0^{PERF} \\ \beta_0^{USE} \end{bmatrix} + \begin{bmatrix} e_i^{PERF} \\ e_i^{USE} \end{bmatrix}$$
$$\mathbf{Y}_i = \mathbf{X}_i \mathbf{B} + \mathbf{e}_i$$

- Using expected values and linear combination rules, we can show that:

$$\mathbf{Y}_i \sim N_2(\mathbf{X}_i \mathbf{B}, \mathbf{V}_i)$$
$$\begin{bmatrix} PERF_i \\ USE_i \end{bmatrix} \sim N_2 \left(\boldsymbol{\mu}_i = \begin{bmatrix} \beta_0^{PERF} \\ \beta_0^{USE} \end{bmatrix}, \mathbf{V}_i = \mathbf{R} = \begin{bmatrix} \sigma_{e:PERF}^2 & \sigma_{e:PERF,USE} \\ \sigma_{e:PERF,USE} & \sigma_{e:USE}^2 \end{bmatrix} \right)$$

$$\begin{bmatrix} PERF_i \\ USE_i \end{bmatrix} \sim N_2 \left(\begin{bmatrix} \beta_0^{PERF} \\ \beta_0^{USE} \end{bmatrix}, \begin{bmatrix} \sigma_{e:PERF}^2 & \sigma_{e:PERF,USE} \\ \sigma_{e:PERF,USE} & \sigma_{e:USE}^2 \end{bmatrix} \right)$$

```
model02.syntax = "
```

```
#Variances:
```

```
perf ~ perf
```

```
use ~ use
```

```
#Covariance:
```

```
perf ~ use
```

```
#Means:
```

```
perf ~ 1
```

```
use ~ 1
```

```
"
```

```
■
```

 $\sigma_{e:PERF}^2$
 $\sigma_{e:USE}^2$
 $\sigma_{e:PERF,USE}$
 β_0^{PERF}
 β_0^{USE}

This covariance matrix is said to be **saturated**: All parameters are estimated

It is also called an **unstructured** covariance matrix

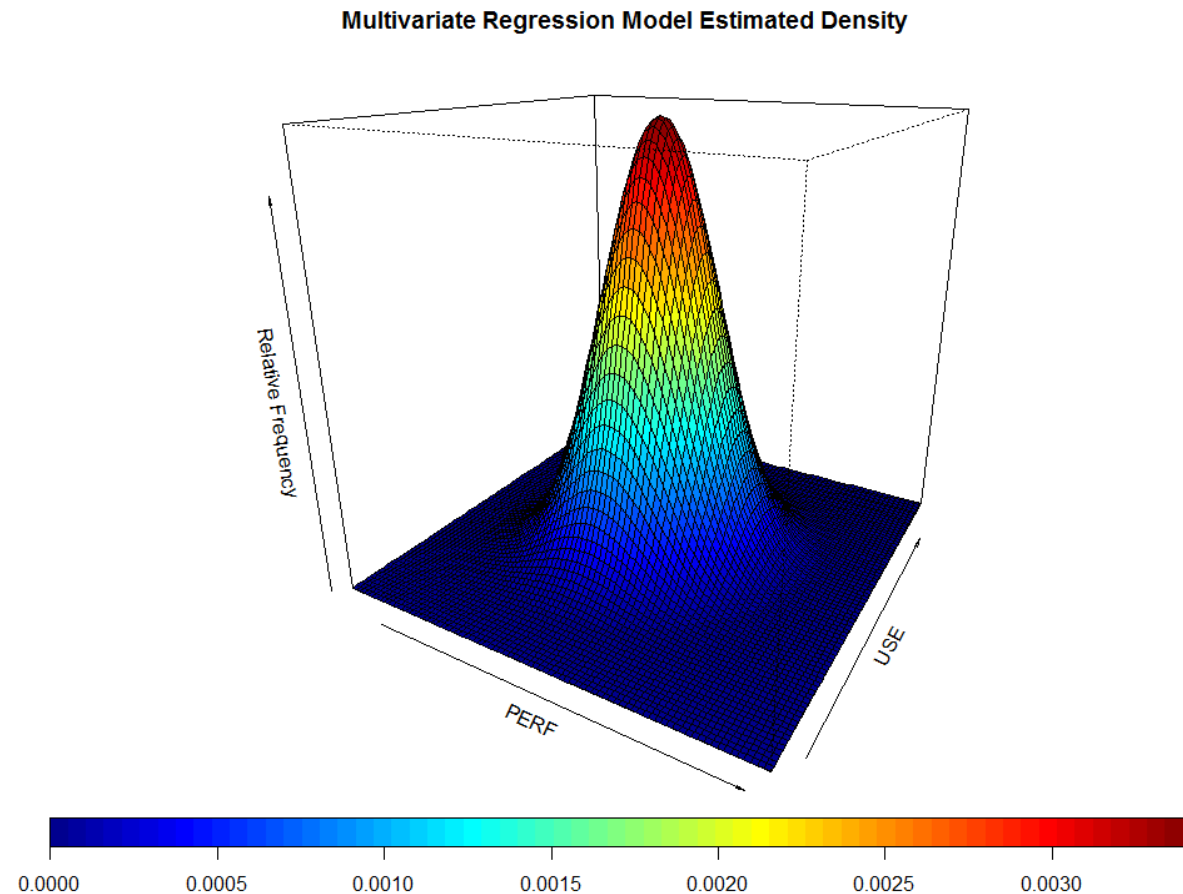
No other structure for the covariance matrix can fit better (only as well as)

➤ The estimated values:

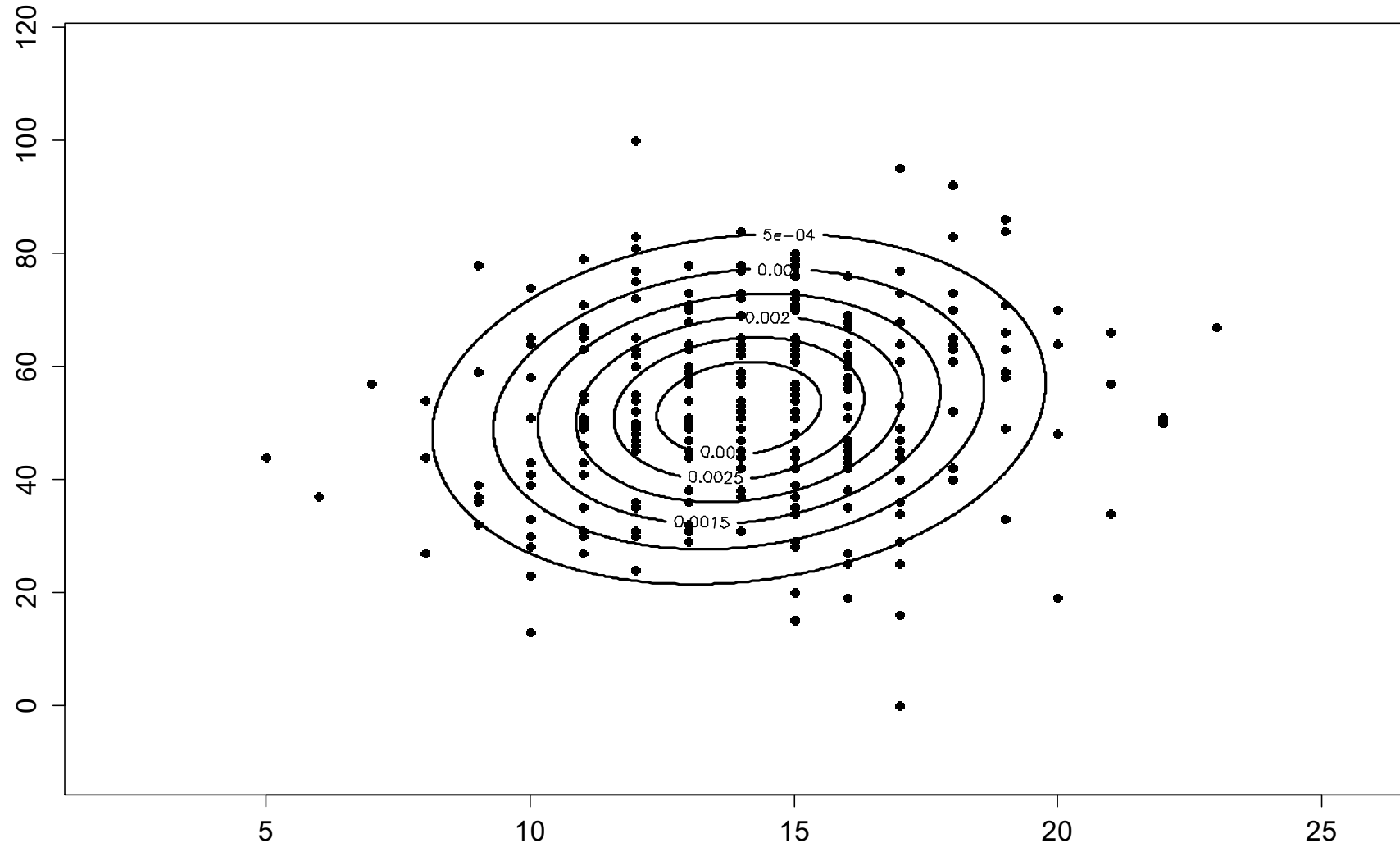
➤ What is the estimated correlation between PERF and USE?

	Estimate	Std.err	Z-value	P(> z)
Covariances:				
perf ~ use	6.847	2.850	2.403	0.016
Intercepts:				
perf	13.959	0.174	80.442	0.000
use	52.440	0.872	60.140	0.000
Variances:				
perf	8.742	0.754	11.596	0.000
use	249.245	19.212	12.973	0.000

$$\begin{bmatrix} PERF_i \\ USE_i \end{bmatrix} \sim N_2 \left(\begin{bmatrix} 13.959 \\ 52.440 \end{bmatrix}, \begin{bmatrix} 8.742 & 6.847 \\ 6.847 & 249.245 \end{bmatrix} \right)$$



Multivariate Regression Model Estimated Density with Data



METHODS OF EXAMINING MODEL FIT

- Model-data fit is of utmost concern when building models with multivariate outcomes
- If a model does not fit the data:
 - Parameter estimates may be biased
 - Standard errors of estimates may be biased
 - Inferences made from the model may be wrong
- Examining model fit is the first step in multivariate models
- That said, not all “good-fitting” models are useful...
 - ...model fit just allows you to talk about your model...there may be nothing of significance (statistically or practically) in your results, though

- Model fit information for models where outcomes are conditionally MVN* come in several types, but all are based on the premise that any model mean and covariance structure must fit as well as the saturated mean vector and covariance matrix model
 - *If model outcomes are not conditionally MVN, model fit is very different**
- All possible models/structures **are nested within** the saturated mean vector and covariance matrix model
 - Most model fit statistics come from comparing any model/structure with the saturated model
- Indices shown first are called “global” model fit indices
 - Report fit of model globally (as opposed to locally for specific parameters)

- We will evaluate the model fit of four models that change some assumptions with our empty multivariate model:

Model #	Mean Vector Structure	Mean Vector Estimates	Covariance Matrix Structure	Covariance Matrix Estimates
00	Saturated	$\begin{bmatrix} 13.959 \\ 52.440 \end{bmatrix}$	Saturated	$\begin{bmatrix} 8.742 & 6.847 \\ 6.847 & 249.245 \end{bmatrix}$
01	Saturated	$\begin{bmatrix} 13.959 \\ 52.440 \end{bmatrix}$	Variance Components	$\begin{bmatrix} 136.002 & 0 \\ 0 & 136.002 \end{bmatrix}$
02	Saturated	$\begin{bmatrix} 13.959 \\ 52.440 \end{bmatrix}$	Independent Variables	$\begin{bmatrix} 8.751 & 0 \\ 0 & 249.201 \end{bmatrix}$
03	Saturated	$\begin{bmatrix} 13.959 \\ 52.440 \end{bmatrix}$	Compound Symmetry	$\begin{bmatrix} 136.004 & 7.207 \\ 7.207 & 136.004 \end{bmatrix}$

- Note: most structural equation models have saturated mean vectors (often not thought of for model fit)
- All model fit information is contained in lavaan output when using the `summary()` function with the `fit.measures = TRUE` option

```
#display empty model output
summary(model02.fit, fit.measures=TRUE)
```

Example lavaan Model Fit Output

```
> summary(model03.fit, fit.measures=TRUE)
lavaan 0.6-19 ended normally after 22 iterations

Estimator              ML
Optimization method    NLMINB
Number of model parameters      5
Number of equality constraints    1

Number of observations      Used      Total
Number of missing patterns      3         350

Model Test User Model:

Test Statistic      Standard      Scaled
Degrees of freedom      1          1
P-value (Chi-square)    0.000      0.000
Scaling correction factor      1.358
Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

Test statistic      6.064      5.573
Degrees of freedom      1          1
P-value      0.014      0.018
Scaling correction factor      1.088

User Model versus Baseline Model:

Comparative Fit Index (CFI)      0.000      0.000
Tucker-Lewis Index (TLI)      -117.981      -95.998

Robust Comparative Fit Index (CFI)      0.000
Robust Tucker-Lewis Index (TLI)      -112.245

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)      -2386.796      -2386.796
Scaling correction factor
for the MLR correction      0.756
Loglikelihood unrestricted model (H1)      -2085.032      -2085.032
Scaling correction factor
for the MLR correction      1.028

Akaike (AIC)      4781.591      4781.591
Bayesian (BIC)      4797.000      4797.000
Sample-size adjusted Bayesian (SABIC)      4784.311      4784.311

Root Mean Square Error of Approximation:

RMSEA      1.316      1.129
90 Percent confidence interval - lower      1.229      1.054
90 Percent confidence interval - upper      1.405      1.206
P-value H_0: RMSEA <= 0.050      0.000      0.000
P-value H_0: RMSEA >= 0.080      1.000      1.000

Robust RMSEA      1.432
90 Percent confidence interval - lower      1.344
90 Percent confidence interval - upper      1.523
P-value H_0: Robust RMSEA <= 0.050      0.000
P-value H_0: Robust RMSEA >= 0.080      1.000

Standardized Root Mean Square Residual:

SRMR      6.723      6.723
```

- **Unlabeled section**
 - **Likelihood ratio test versus the saturated model**
 - **Testing if your model fits as well as the saturated model**
- **Model test baseline model**
 - Likelihood ratio test pitting the saturated model against the independent variables model
 - Testing whether any variables have non-zero covariances (significant correlations)
- **User model versus baseline model**
 - CFI
 - TLI
- **Loglikelihood and Information Criteria**
 - Likelihood ratio tests (nested models)
 - Information criteria comparisons (non-nested models)
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 - How far off a model's correlations are from the saturated model correlations

- Primary: obtained model χ^2 (from Model test baseline model)
 - here we use the MLR rescaled χ^2 from the “Robust” Column
- χ^2 is evaluated based on model df (difference in parameters between your CFA model and the saturated model)
- Tests null hypothesis that **this** model (H_0) fits equally to **saturated model** (H_1) so significance is undesirable (smaller χ^2 , bigger p-value is better)
 - Means saturated model is estimated **automatically** for each model analyzed
- Just using χ^2 is insufficient, however:
 - Distribution doesn't behave like a true χ^2 if sample sizes are small (or, if not using MLR, if items are non-normally distributed)
 - Obtained χ^2 depends largely on sample size
 - Some mention this is an unreasonable null hypothesis (perfect fit??)
- Because of these issues, alternative measures of fit are usually used in conjunction with the χ^2 test of model fit
 - Absolute Fit Indices (besides χ^2)
 - Parsimony-Corrected; Comparative (Incremental) Fit Indices

- The Chi-Square Test of Model Fit provides a likelihood ratio test comparing the current model to the **saturated (unstructured) model**:
 - The value is -2 times the difference in log-likelihoods (rescaled if MLR)
 - The degrees of freedom is the difference in the number of estimated model parameters
 - The p-value is from the Chi-square distribution
- **If this test has a significant p-value:**
 - The current model (H_0) is rejected – the model fit is significantly worse than the full model
 - In latent variable models, this test is usually ignored
 - Said to be overly sensitive
- **If this test does not have a significant p-value:**
 - The current model (H_0) is not rejected – **fits equivalently to full model**

➤ Model 00 (saturated covariance matrix):

Estimator	ML	Robust
Minimum Function Test Statistic	0.000	0.000
Degrees of freedom	0	0
Minimum Function Value	0.00000000000000	
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		NA

➤ Model 01 (variance components covariance matrix):

Estimator	ML	Robust
Minimum Function Test Statistic	604.272	804.743
Degrees of freedom	2	2
P-value (Chi-square)	0.000	0.000
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		0.751

➤ Model 02 (independent variables covariance matrix):

Estimator	ML	Robust
Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value (Chi-square)	0.014	0.018
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.088

➤ Model 03 (compound symmetry covariance matrix):

Estimator	ML	Robust
Minimum Function Test Statistic	603.527	444.560
Degrees of freedom	1	1
P-value (Chi-square)	0.000	0.000
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.358

- The saturated model LRT comes from a likelihood ratio test of the current model with the saturated model
- If using MLR (Robust method), then this LRT is rescaled based on the estimated scaling factors of both models
- This same information can be obtained from:
 - Loglikelihood model output section
 - `anova()` function comparing fit for current and saturated models

```
> # likelihood ratio tests:
> anova(model00.fit, model01.fit)

Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

lavaan->lavTestLRT():
lavaan NOTE: The "Chisq" column contains standard test statistics, not the robust
test that should be reported per model. A robust difference test is a function of two
standard (not robust) statistics.
      Df    AIC    BIC Chisq Chisq diff Df diff Pr(>Chisq)
model00.fit  0 4180.1 4199.3  0.00
model01.fit  2 4780.3 4791.9 604.27    804.74      2 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova(model00.fit, model02.fit)

Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

lavaan->lavTestLRT():
lavaan NOTE: The "Chisq" column contains standard test statistics, not the robust
test that should be reported per model. A robust difference test is a function of two
standard (not robust) statistics.
      Df    AIC    BIC Chisq Chisq diff Df diff Pr(>Chisq)
model00.fit  0 4180.1 4199.3  0.0000
model02.fit  1 4184.1 4199.5  6.0641     5.5729      1  0.01824 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> anova(model00.fit, model03.fit)

Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

lavaan->lavTestLRT():
lavaan NOTE: The "Chisq" column contains standard test statistics, not the robust
test that should be reported per model. A robust difference test is a function of two
standard (not robust) statistics.
      Df    AIC    BIC Chisq Chisq diff Df diff Pr(>Chisq)
model00.fit  0 4180.1 4199.3  0.00
model03.fit  1 4781.6 4797.0 603.53    444.56      1 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


➤ From the lavaan output:

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-2088.064	-2088.064
Scaling correction factor for the MLR correction		1.012
Loglikelihood unrestricted model (H1)	-2085.032	-2085.032
Scaling correction factor for the MLR correction		1.028
Number of free parameters	4	4

➤ Calculation:

➤ 4 parameters in our model; 5 in saturated model

➤ Scaling correction factor:

$$c_{LR} = \left| \frac{(q_{restricted})(c_{restricted}) - (q_{full})(c_{full})}{(q_{restricted} - q_{full})} \right| = \left| \frac{(4 \times 1.012) - (5 \times 1.028)}{4 - 5} \right| \approx 1.088$$

➤ $\chi^2 = -\frac{2^*(-2,088.064 - -2,085.032)}{1.088} = \frac{6.064}{1} = 5.573$

➤ DF = 1

Estimator	ML	Robust
Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value (Chi-square)	0.014	0.018
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.088

- Look at the following output from the saturated model (model 00):

```
Estimator                                ML      Robust
Minimum Function Test Statistic          0.000    0.000
Degrees of freedom                        0         0
Minimum Function Value                   0.000000000000000
Scaling correction factor                NA
  for the Yuan-Bentler correction (Mplus variant)

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)            -2085.032  -2085.032
Loglikelihood unrestricted model (H1)     -2085.032  -2085.032

Number of free parameters                 5         5
```

- If the loglikelihoods of the current model (“User model” or H_0) are equal to the loglikelihoods of the saturated model (“Unrestricted model” or H_1), then you are running a model that is equivalent to the saturated model
 - **No other model fit will be available or useful**

- ~~Unlabeled section~~
 - ~~Likelihood ratio test versus the saturated model~~
 - ~~Testing if your model fits as well as the saturated model~~
- **Model test baseline model**
 - **Likelihood ratio test pitting the saturated model against the independent variables model**
 - **Testing whether any variables have non-zero covariances (significant correlations)**
- **User model versus baseline model**
 - CFI
 - TLI
- **Loglikelihood and Information Criteria**
 - Likelihood ratio tests (nested models)
 - Information criteria comparisons (non-nested models)
- **Root Mean Square Error of Approximation**
 - How far off a model is from the saturated model, per degree of freedom
- **Standardized Root Mean Square Residual**
 - How far off a model's correlations are from the saturated model correlations

- The “model test baseline model” section provides a LRT:
 - Comparing the saturated (unstructured) model (MODEL 00) with an independent variables model (called the baseline model) (MODEL 03)
 - Note: this is equal to previous LRT in Model 03 output:

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value	0.014	0.018

- Here, the “null” model is the baseline (the independent variables model)
 - If the test is significant, this means that at least one (and likely more than one) variable has a significant covariance (and correlation)
 - If the test is not significant, this means that the independence model is appropriate
 - This is not likely to happen
 - But if it does, there are virtually no other models that will be significant
- Not often reported as it is likely variables are correlated

➤ Model 00 (saturated covariance matrix):

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value	0.014	0.018

➤ Model 01 (variance components covariance matrix):

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value	0.014	0.018

➤ Model 02 (independent variables covariance matrix):

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value	0.014	0.018

➤ Model 04 (compound symmetry covariance matrix):

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value	0.014	0.018

- ~~Unlabeled section~~
 - ~~Likelihood ratio test versus the saturated model~~
 - ~~Testing if your model fits as well as the saturated model~~

- ~~Model test baseline model~~
 - ~~Likelihood ratio test pitting the saturated model against the independent variables model~~
 - ~~Testing whether any variables have non-zero covariances (significant correlations)~~

- **User model versus baseline model**
 - **CFI**
 - **TLI**

- **Loglikelihood and Information Criteria**
 - Likelihood ratio tests (nested models)
 - Information criteria comparisons (non-nested models)

- **Root Mean Square Error of Approximation**
 - How far off a model is from the saturated model, per degree of freedom

- **Standardized Root Mean Square Residual**
 - How far off a model's correlations are from the saturated model correlations

- The “User model versus baseline model” section provides two additional measures of model fit comparing the current (user) model to the baseline (independent variables) model

User model versus baseline model:

Comparative Fit Index (CFI)	1.000	1.000
Tucker-Lewis Index (TLI)	1.000	1.000

- CFI stands for Comparative Fit Index
 - Higher is better (above .95 indicates good fit)
- TLI stands for Tucker Lewis Index
 - Higher is better (above .95 indicates good fit)

- Fit evaluated relative to a ‘null’ model (of 0 covariances)

- Relative to that, your model should be great!

T = target (current/estimated) model

N = null (baseline/independent variables) model

- **CFI: Comparative Fit Index**

- Based on idea of the chi-square non-centrality parameter: $(\chi^2 - df)$

- $$CFI = 1 - \frac{\max(\chi_T^2 - df_T, 0)}{\max(\chi_T^2 - df_T, \chi_N^2 - df_N, 0)}$$

- From 0 to 1: bigger is better, $> .90$ = “acceptable”, $> .95$ = “good”

- **TLI: Tucker-Lewis Index (= Non-Normed Fit Index)**

- $$TLI = \frac{\frac{\chi_N^2}{df_N} - \frac{\chi_T^2}{df_T}}{\frac{\chi_N^2}{df_N} - 1}$$

- From <0 to >1 , bigger is better, $> .95$ = “good”

The estimated model
(Model 02; the target)

Estimator	ML	Robust
Minimum Function Test Statistic	0.000	0.000
Degrees of freedom	0	0
Minimum Function Value	0.00000000000000	
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		NA

The independent variables model
(Model 04; the null model)

Estimator	ML	Robust
Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value (Chi-square)	0.014	0.018
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.088

Compute numerator:

$$\max(\chi_T^2 - df_T, 0) = \max(0 - 0, 0) = 0$$

Compute denominator:

$$\begin{aligned} &\max(\chi_T^2 - df_T, \chi_N^2 - df_N, 0) \\ &= \max(0, 5.573 - 1, 0) = 4.573 \end{aligned}$$

Compute CFI:

$$\begin{aligned} CFI &= 1 - \frac{\max(\chi_T^2 - df_T, 0)}{\max(\chi_T^2 - df_T, \chi_N^2 - df_N, 0)} = 1 - \frac{0}{4.573} \\ &= 1.000 \end{aligned}$$

User model versus baseline model:

Comparative Fit Index (CFI)	1.000	1.000
Tucker-Lewis Index (TLI)	1.000	1.000

Tucker-Lewis Index Calculation

The estimated model
(Model 02; the target)

Estimator	ML	Robust
Minimum Function Test Statistic	0.000	0.000
Degrees of freedom	0	0
Minimum Function Value	0.00000000000000	
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		NA

Compute Target Model ratio:

$$\frac{\chi_T^2}{df_T} = \frac{0}{0} = ?$$

The independent variables model
(Model 04; the null model)

Estimator	ML	Robust
Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value (Chi-square)	0.014	0.018
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.088

Compute Null Model Ratio:

$$\frac{\chi_N^2}{df_N} = \frac{5.573}{1} = 5.573$$

Compute TLI:

$$TLI = \frac{\frac{\chi_N^2}{df_N} - \frac{\chi_T^2}{df_T}}{\frac{\chi_N^2}{df_N} - 1} = \frac{5.573 - ?}{5.573 - 1} = ?$$

User model versus baseline model:

Comparative Fit Index (CFI)	1.000	1.000
Tucker-Lewis Index (TLI)	1.000	1.000

- The information criteria output provides relative fit statistics:

```
Information Criteria

Akaike (AIC)                7172.878
Bayesian (BIC)              7223.031
Sample-Size Adjusted BIC    7181.790
(n* = (n + 2) / 24)
```

- AIC: Akaike Information Criterion
 - BIC: Bayesian Information Criterion (also called Schwarz's criterion)
 - Sample-size Adjusted BIC
-
- These statistics weight the information given by the parameter values by the parsimony of the model (the number of model parameters)
 - For all statistics, the smaller number is better
 - The core of these statistics is $-2 \cdot \log\text{-likelihood}$

- ~~Unlabeled section~~
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 - ~~Testing whether any variables have non-zero covariances (significant correlations)~~
- ~~User model versus baseline model~~
 - ~~CFI~~
 - ~~TLI~~
- **Loglikelihood and Information Criteria**
 - **Likelihood ratio tests (nested models)**
 - **Information criteria comparisons (non-nested models)**
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- **Standardized Root Mean Square Residual**
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- Information criteria are relative tests of fit

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-2085.032	-2085.032
Loglikelihood unrestricted model (H1)	-2085.032	-2085.032
Number of free parameters	5	5
Akaike (AIC)	4180.064	4180.064
Bayesian (BIC)	4199.325	4199.325
Sample-size adjusted Bayesian (BIC)	4183.464	4183.464

- They are calculated based on the log-likelihood of the model, factoring in a penalty for number of parameters (plus other things)
- They should never be used to compare nested models
 - The likelihood ratio test is the most powerful test statistic to use for nested models
- When comparing non-nested models, first choose a statistic
 - AIC, BIC, or Sample-size Adjusted BIC are what are given by default
- The preferred model is the one with the lowest value of that statistic

- ~~Unlabeled section~~
 - ~~Likelihood ratio test versus the saturated model~~
 - ~~Testing if your model fits as well as the saturated model~~
- ~~Model test baseline model~~
 - ~~Likelihood ratio test pitting the saturated model against the independent variables model~~
 - ~~Testing whether any variables have non-zero covariances (significant correlations)~~
- ~~User model versus baseline model~~
 - ~~CFI~~
 - ~~TLI~~
- ~~Loglikelihood and Information Criteria~~
 - ~~Likelihood ratio tests (nested models)~~
 - ~~Information criteria comparisons (non-nested models)~~
- **Root Mean Square Error of Approximation**
 - **How far off a model is from the saturated model, per degree of freedom**
- **Standardized Root Mean Square Residual**
 - **How far off a model's correlations are from the saturated model correlations**

Parsimony-Corrected: **RMSEA**

- **Root Mean Square Error of Approximation**
- Uses comparison with CFA model and saturated model
 - χ^2 listed here from first part of lavaan output
- Relies on a non-centrality parameter (NCP)
 - Indexes how far off your model is $\rightarrow \chi^2$ distribution shoved over
 - $NCP \rightarrow d = (\chi^2 - df) / (N-1)$ Then, $RMSEA = \text{SQRT}(d/df)$
 - df is difference between # parameters in CFA model and saturated model
 - RMSEA ranges from 0 to 1; smaller is better
 - $< .05$ or $.06$ = “good”, $.05$ to $.08$ = “acceptable”,
 $.08$ to $.10$ = “mediocre”, and $> .10$ = “unacceptable”
 - In addition to point estimate, get 90% confidence interval
 - RMSEA penalizes for model complexity – it’s discrepancy in fit per df left in model (but not sensitive to N , although CI can be)
 - Test of “close fit”: null hypothesis that $RMSEA \leq .05$

- The RMSEA is an index of model fit where 0 indicates perfect fit (smaller is better; model 04 shown):

Root Mean Square Error of Approximation:

RMSEA		0.121	0.115
90 Percent Confidence Interval	0.044	0.220	0.040 0.210
P-value RMSEA $\leq$ 0.05		0.063	0.073

- RMSEA is based on the approximated covariance matrix
 - More on this in two weeks
- The goal is a model with an RMSEA less than .05
 - Although there is some flexibility
- The result above indicates our model fits well (RMSEA of .026)
 - Expected for 13 parameters (out of 14 possible)

➤ From lavaan:

Estimator	ML	Robust
Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value (Chi-square)	0.014	0.018
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.088
Number of observations		Used 348

$$\text{Create Non-Centrality Parameter } d = \frac{\chi^2 - df}{N - 1} = \frac{5.573 - 1}{348 - 1} = 0.013$$

Calculate RMSEA:

$$\text{RMSEA} = \sqrt{\frac{d}{df}} = \sqrt{\frac{0.013}{1}} = 0.115$$

Root Mean Square Error of Approximation:

RMSEA		0.121	0.115
90 Percent Confidence Interval	0.044	0.220	0.040 0.210
P-value RMSEA <= 0.05		0.063	0.073

- ~~Unlabeled section~~
 - ~~Likelihood ratio test versus the saturated model~~
 - ~~Testing if your model fits as well as the saturated model~~

- ~~Model test baseline model~~
 - ~~Likelihood ratio test pitting the saturated model against the independent variables model~~
 - ~~Testing whether any variables have non-zero covariances (significant correlations)~~

- ~~User model versus baseline model~~
 - ~~CFI~~
 - ~~TLI~~

- ~~Loglikelihood and Information Criteria~~
 - ~~Likelihood ratio tests (nested models)~~
 - ~~Information criteria comparisons (non-nested models)~~

- ~~Root Mean Square Error of Approximation~~
 - ~~How far off a model is from the saturated model, per degree of freedom~~

- **Standardized Root Mean Square Residual**
 - **How far off a model's correlations are from the saturated model correlations**

- The SRMR (standardized root mean square residual) provides the average standardized difference between:
 - The estimated covariance matrix of the saturated model
 - The estimated covariance matrix of the current model

Standardized Root Mean Square Residual:

SRMR

0.066

0.066

- The model-estimated covariance matrix for model 04:

```
> model04.estimate$cov
      perf      use
perf  8.751
use   0.000 249.201
```

- The model estimated covariance matrix for model 02:

```
> model02.estimate$cov
      perf      use
perf  8.742
use   6.847 249.245
> |
```

- Lower is better (some suggest less than 0.08)

- For our example, we have four models
 - One is the saturated model which is guaranteed to fit

- Therefore, we have to ask the following set of questions:
 - Does model 01 fit?
 - Does model 02 fit?
 - Does model 03 fit?

- Of those that fit, which is preferred?

```
> summary(model03.fit, fit.measures=TRUE)
lavaan (0.5-17) converged normally after 6 iterations
```

	Used	Total	
Number of observations	348	350	
Number of missing patterns	3		
Estimator	ML	Robust	
Minimum Function Test Statistic	604.272	804.743	
Degrees of freedom	2	2	
P-value (Chi-square)	0.000	0.000	
Scaling correction factor		0.751	
for the Yuan-Bentler correction (Mplus variant)			

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573	
Degrees of freedom	1	1	
P-value	0.014	0.018	

User model versus baseline model:

Comparative Fit Index (CFI)	0.000	0.000	
Tucker-Lewis Index (TLI)	-58.465	-86.772	

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-2387.168	-2387.168	
Scaling correction factor		1.212	
for the MLR correction			
Loglikelihood unrestricted model (H1)	-2085.032	-2085.032	
Scaling correction factor		1.028	
for the MLR correction			
Number of free parameters	3	3	
Akaike (AIC)	4780.336	4780.336	
Bayesian (BIC)	4791.893	4791.893	
Sample-size adjusted Bayesian (BIC)	4782.376	4782.376	

Root Mean Square Error of Approximation:

RMSEA		0.930	1.074	
90 Percent Confidence Interval	0.869	0.993	1.003	1.147
P-value RMSEA <= 0.05		0.000	0.000	

Standardized Root Mean Square Residual:

SRMR	6.723	6.723	
------	-------	-------	--

```
> summary(model04.fit, fit.measures=TRUE)
lavaan (0.5-17) converged normally after 18 iterations
```

Number of observations	Used 348	Total 350	
Number of missing patterns	3		
Estimator	ML	Robust	
Minimum Function Test Statistic	6.064	5.573	
Degrees of freedom	1	1	
P-value (Chi-square)	0.014	0.018	
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.088	

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573	
Degrees of freedom	1	1	
P-value	0.014	0.018	

User model versus baseline model:

Comparative Fit Index (CFI)	0.000	0.000	
Tucker-Lewis Index (TLI)	-0.000	-0.000	

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-2088.064	-2088.064	
Scaling correction factor for the MLR correction		1.012	
Loglikelihood unrestricted model (H1)	-2085.032	-2085.032	
Scaling correction factor for the MLR correction		1.028	
Number of free parameters	4	4	
Akaike (AIC)	4184.128	4184.128	
Bayesian (BIC)	4199.537	4199.537	
Sample-size adjusted Bayesian (BIC)	4186.848	4186.848	

Root Mean Square Error of Approximation:

RMSEA	0.121	0.115	
90 Percent Confidence Interval	0.044 0.220	0.040 0.210	
P-value RMSEA <= 0.05	0.063	0.073	

Standardized Root Mean Square Residual:

SRMR	0.066	0.066	
------	-------	-------	--

```
> summary(model05.fit, fit.measures=TRUE)
lavaan (0.5-17) converged normally after 21 iterations
```

	Used	Total	
Number of observations	348	350	
Number of missing patterns	3		
Estimator	ML	Robust	
Minimum Function Test Statistic	603.527	444.560	
Degrees of freedom	1	1	
P-value (Chi-square)	0.000	0.000	
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.358	

Model test baseline model:

Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value	0.014	0.018

User model versus baseline model:

Comparative Fit Index (CFI)	0.000	0.000
Tucker-Lewis Index (TLI)	-117.981	-95.998

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-2386.796	-2386.796
Scaling correction factor for the MLR correction		0.945
Loglikelihood unrestricted model (H1)	-2085.032	-2085.032
Scaling correction factor for the MLR correction		1.028
Number of free parameters	4	4
Akaike (AIC)	4781.591	4781.591
Bayesian (BIC)	4797.000	4797.000
Sample-size adjusted Bayesian (BIC)	4784.311	4784.311

Root Mean Square Error of Approximation:

RMSEA	1.316	1.129	
90 Percent Confidence Interval	1.229 1.405	1.054	1.206
P-value RMSEA <= 0.05	0.000	0.000	

Standardized Root Mean Square Residual:

SRMR	6.723	6.723
------	-------	-------

LOCAL MODEL FIT MEASURES

- Local measures of model (mis)fit are statistics that point to the location (typically of a covariance matrix) where a model may not fit well
 - As opposed to “global” measures that indicate a model fit overall
- Local measures of model (mis)fit are typically of two types:
 - Residual covariance matrices (unstandardized, standardized, or normalized)
 - The difference between the model’s estimated covariance matrix and the saturated model’s estimated covariance matrix
 - These were used for the SRMR
 - Model “modification indices”
 - 1-degree of freedom hypothesis tests for the improvement of the model LRT if one more parameter was allowed to be estimated

- The model estimated covariance matrices for model 02:

```
> model02.estimated$cov
```

	perf	use
perf	8.751	
use	0.000	249.201

- The “raw” or “unstandardized” residual covariance matrix for model 04 (literally taking model04 – model02):

```
> model02.residuals = residuals(model02.fit, type="raw")
> model02.residuals$cov
```

	perf	use
perf	-0.009	
use	6.847	0.044

- Shows that the biggest difference is the covariance between PERF and USE

- I often prefer “normalized” versions of these matrices

- We can inspect the normalized residual covariance matrix (like z-scores) to see where our biggest misfit occurs

```
> residuals(model02.fit, type="normalized")$cov
```

	perf	use
perf	-0.012	
use	2.403	0.002

- This indicates we should estimate the covariance (> 1.96)

- As we used Maximum Likelihood to estimate our model, another useful feature is that of the modification indices
 - Modification indices, also called Score or LaGrangian Multiplier tests, attempt to suggest the change in the log-likelihood for adding a given model parameter (larger values indicate a better fit for adding the parameter)

```
> # modification indices  
> modindices(model02.fit)  
      lhs op rhs      mi      epc sepc.lv sepc.all sepc.nox  
3 perf ~~ use 6.239 7.125    7.125    0.153    0.153
```

- mi column: the expected value of the LRT of the current model (model 04) and a model where this parameter was added
- mi.scaled column: the scaled (robust) LRT
 - Should be bigger than 3.84 for 1 df
 - Practice is to find values that are much higher (say 10 or more)
- epc column: expected value of the parameter in the model where this parameter was added

- As the covariance parameter was suggested to be added by the MI, here is what happens when we add the parameter and re-estimate the model

```

> # model re-estimation
> modindices(model02.fit)
      lhs op rhs      mi      epc sepc.lv sepc.all sepc.nox
3 perf ~~ use 6.239 7.125    7.125    0.153    0.153

```

- By adding that parameter, we get model 00:

Estimator	ML	Robust
Minimum Function Test Statistic	6.064	5.573
Degrees of freedom	1	1
P-value (Chi-square)	0.014	0.018
Scaling correction factor		1.088
for the Yuan-Bentler correction (Mplus variant)		

LRT between
Model 00 and Model 02

	Estimate	Std.err	Z-value	P(> z)
Covariances:				
perf ~~				
use	6.847	2.850	2.403	0.016

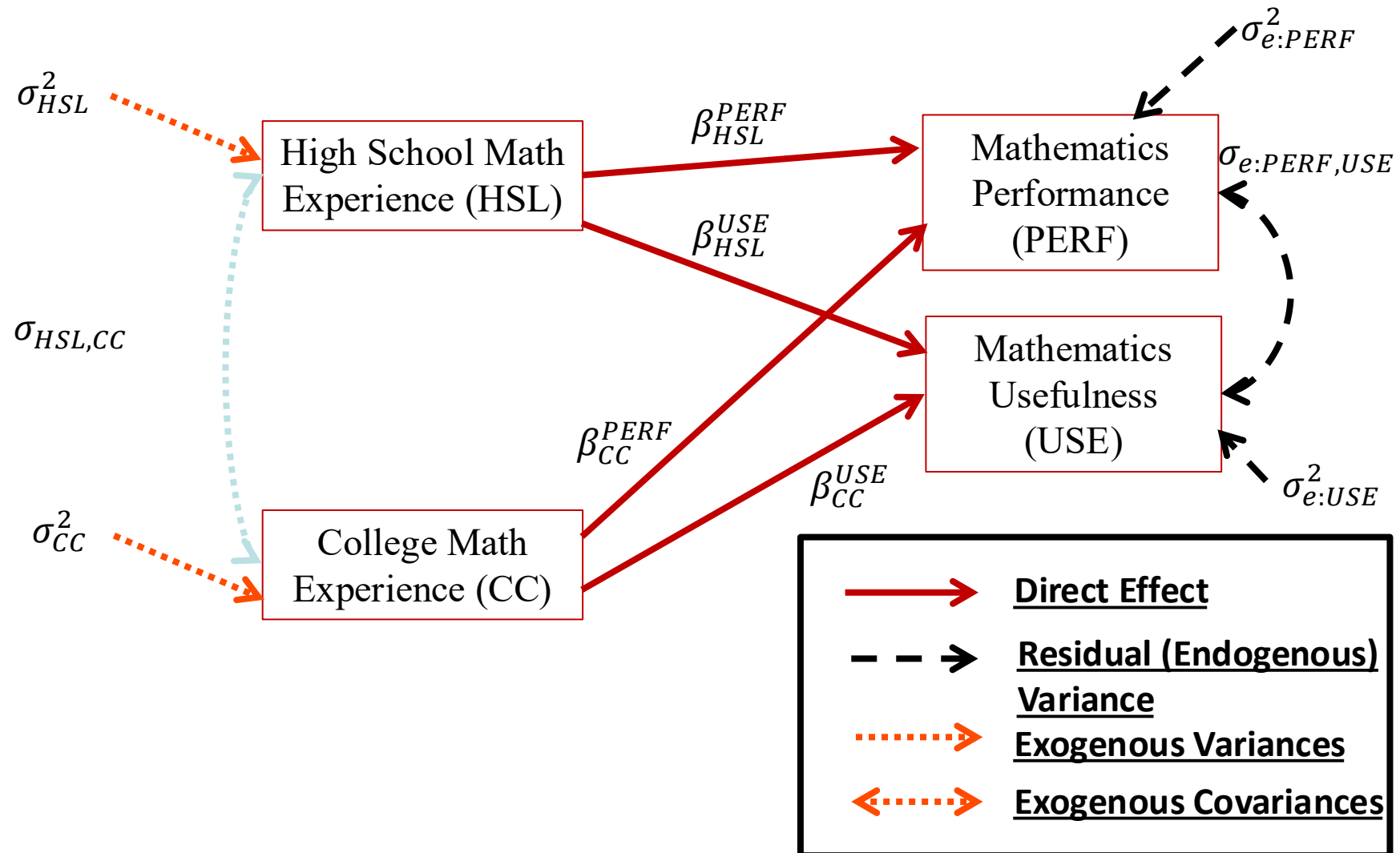
Estimated Covariance Parameter

- Next will add predictors to both variables at once:
 - Predicting mathematics performance (PERF) with high school (HSL) and college (CC) experience
 - Predicting perceived usefulness (USE) with high school (HSL) and College (CC) experience

$$\begin{aligned} PERF_i &= \beta_0^{PERF} + \beta_{HSL}^{PERF} HSL_i + \beta_{CC}^{PERF} CC_i + e_i^{PERF} \\ USE_i &= \beta_0^{USE} + \beta_{HSL}^{USE} HSL_i + \beta_{CC}^{USE} CC_i + e_i^{USE} \end{aligned}$$

- We denote the residual for PERF as e_i^{PERF} and the residual for USE as e_i^{USE}
 - Here, we assume the residuals are Multivariate Normal:

$$\begin{bmatrix} e_i^{PERF} \\ e_i^{USE} \end{bmatrix} \sim N_2 \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{e:PERF}^2 & \sigma_{e:PERF,USE} \\ \sigma_{e:PERF,USE} & \sigma_{e:USE}^2 \end{bmatrix} \right)$$



- An important distinction in path analysis is between endogenous and exogenous variables
- Endogenous variable(s): variables whose variability *is explained* by one or more variables in a model
 - In linear regression, the **dependent variable** is the only endogenous variable in an analysis
 - Mathematics Performance (PERF) in our example
- Exogenous variable(s): variables whose variability *is not explained* by any variables in a model
 - In linear regression, the **independent variable(s)** are the exogenous variables in the analysis
 - High school (HSL) and college (CC) experience

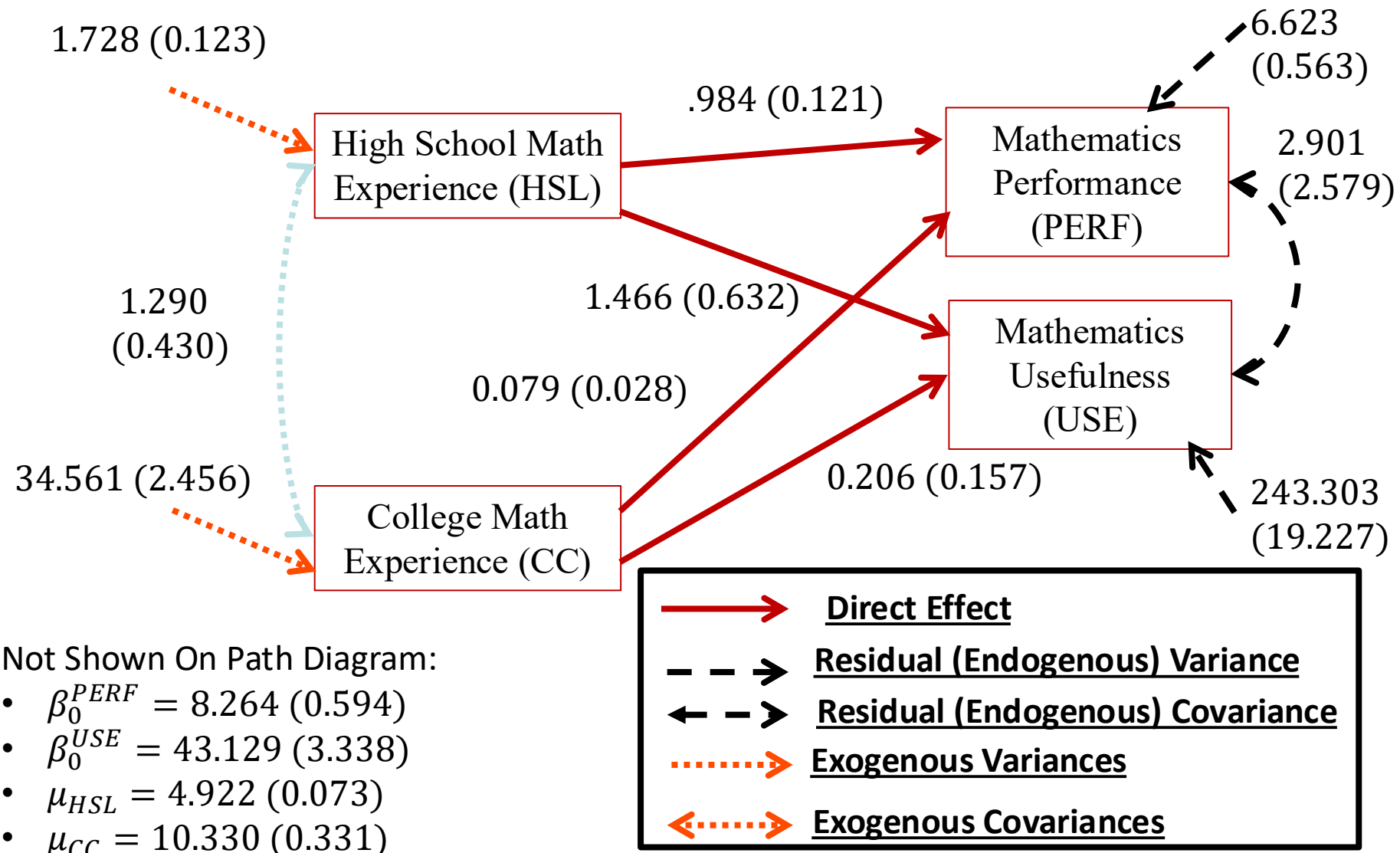
- The endogenous (dependent) variables are:
 - Performance (PERF) and Usefulness (USE)

- The exogenous (independent) variables are:
 - High school (HSL) and college (CC) experience

- Lavaan also puts the exogenous variables into the likelihood function
 - Could be problematic if these are not MVN (like gender)
 - We will ignore this for now
- The `fixed.x = TRUE` option takes the sample means/variances of these variables and plugs them into the likelihood
 - We'll do this explicitly through syntax and turn this option off

```
model05.syntax = "  
#endogenous variable equations  
perf ~ hsl + cc  
use ~ hsl + cc  
  
#endogenous variable intercepts  
perf ~ 1  
use ~ 1  
  
#endogenous variable residual variances  
perf ~~ perf  
use ~~ use  
  
#endogenous variable residual covariances  
perf~~ use  
  
#exogenous variables put into likelihood function:  
  
#exogenous means(intercepts)  
hsl ~ 1  
cc ~ 1  
  
#exogenous variances  
hsl ~~ hsl  
cc ~~ cc  
  
#exogenous covariances  
hsl ~~ cc  
"
```

- If we considered all four variables to be part of a multivariate normal distribution, our unstructured (saturated) model would have 14 parameters:
 - 4 means
 - 4 variances
 - 6 covariances (4-choose-2 or $4*(4-1)/2$)
- The model itself has 14 parameters:
 - 4 intercepts
 - 4 slopes
 - 2 residual variances
 - 1 residual covariance
 - 2 exogenous variances
 - 1 exogenous covariance
- Therefore, this model will fit perfectly – no model fit statistics will be available
 - Even without model fit, interpretation of parameters can proceed



- The scale of the (unstandardized) slope coefficients is given in terms of UNITS of Y (SD Y) per UNITS of X (SD X)
 - Y goes up β_X^Y UNITS of Y for every UNIT of X
 - HSL has SD of 1.31; CC has SD of 5.88
 - If the UNITS of X differ for the various IVs in a model, it can be hard to compare relative strengths of coefficients
 - $\beta_{HSL}^{PERF} = .984$ (but HSL has SD of 1.31)
 - $\beta_{CC}^{PERF} = .079$ (but CC has SD of 5.88)
- Standardized coefficients are the coefficients that would be obtained if Y and X were standardized:
 - Standardized = variance of 1 (i.e. z-scores used for analysis)
- Standardized coefficients are useful for comparing the relative effects of each IV in the model

- To get standardized estimates, add the “standardized=TRUE” option to the summary statement

```
#standardized parameter estimates:  
summary(model01.fit, fit.measures=TRUE, standardized=TRUE)
```

- Under the output section, the last two columns are the standardized estimates (std.lv and std.all)

- Also available from the function standardizedSolution (adds no.x type)

```
standardizedSolution(model01.fit, type = "std.all")
```

- Three types of standardizations are given:

- **std.all**: These are the standardized regression coefficients; use these for continuous IVs (used for our current analysis)
- **std.noX**: These only standardize based on variance of Y (the DV). Use when binary variables are IVs (like gender dummy coding) as unit of X has no meaning
- **std.lv**: Discussed when we get to models with latent variables

- $\beta_0^{PERF} = 8.264$: the intercept for PERF – the value of PERF when all predictors are zero (HSL = 0 and CC = 0)
- $\beta_{HSL}^{PERF} = 0.986$: the slope for HSL predicting PERF. Indicates that for every one-unit increase in HSL (holding CC constant), PERF increases by .986
 - The standardized coefficient was .438
- $\beta_{CC}^{PERF} = 0.079$: the slope for CC predicting PERF. Indicates that for every one-unit increase in CC (holding HSL constant), PERF increases by .079
 - The standardized coefficient was .157

- $\beta_0^{USE} = 43.129$: the intercept for USE – the value of USE when all predictors are zero (HSL = 0 and CC = 0)
- $\beta_{HSL}^{USE} = 1.466$: the slope for HSL predicting USE. Indicates that for every one-unit increase in HSL (holding CC constant), USE increases by 1.466
 - The standardized coefficient was .122
- $\beta_{CC}^{USE} = 0.206$: the slope for CC predicting USE. Indicates that for every one-unit increase in CC (holding HSL constant), USE increases by .206. This was found to be not significant, meaning college experience did not predict perceived usefulness
 - The standardized coefficient was .077

- $\sigma_{e:PERF}^2 = 6.623$: the residual variance for PERF
 - The R^2 for PERF was .240 (the same as before)
- $\sigma_{e:USE}^2 = 243.303$: the residual variance for USE
 - The R^2 for USE was .024 (a very small effect)
- $\sigma_{e:PERF,USE} = 2.901$: the residual covariance between USE and PERF
 - This value was not significant, meaning we can potentially set its value to zero and re-estimate the model
- Each describes the amount of variance not accounted for in each dependent (endogenous) variable

- After adding both independent variables HSL and CC, the residual variance of performance was $\sigma_{e:PERF}^2 = 6.631$
 - Value of PERF's variance estimate from Model 01
- Therefore, the inclusion of these variables reduced the variance of PERF from 8.722 to 6.631, for an

$$R^2 = \frac{8.722 - 6.631}{8.722} = .24$$

- lavaan reports this value by using the inspect() function:

```
> inspect(model01.fit, what="r2")  
perf    use  
0.240 0.024
```

- This lecture introduced multivariate linear models within the structural equation modeling framework
- Although most of the models are very familiar, the biggest difference between a multivariate linear model and a univariate model is that model fit checking is needed in multivariate linear models
 - This is why I hesitate to call the R^2 value from a univariate linear model “model fit”
- The SEM framework is very flexible and incorporates more complex models such as path analyses, latent variable models, and their combinations